Date: / /

41 Ans Cost Price = 650 R.S

Selling Price = 715 R.S

Profit = 715 - 650 = 65 R.S

$P\% = \frac{P}{CP} \times 100 = \frac{65}{650} \times 100$

$= \frac{6500}{650} = 10\%$.

42 Ans Average age of 30 students = 20 years.

⇒ Total age = 30 × 20 = 600 years

and Total age of 20 other students

= 20 × 30 = 600 year.

Total students = 30 + 20 = 50

Total age of 50 students = 600 + 600 = 1200 R.S

and average age = $\frac{1200}{50}$ = 24 years.

43 Selling price = 630 R.S

Profit = 5%

CP = ?

⇒ $CP = \frac{SP \times 100}{100 + P\%} = \frac{630 \times 100}{100 + 5} = \frac{63000}{105}$

= 600 R.S.

44 Ans Formula

Population before n years = $\frac{P}{\left(1 + \frac{R}{100}\right)^n}$

⇒ $\frac{370440}{\left(1 + \frac{5}{100}\right)^3} = \frac{370440}{\frac{21}{20} \times \frac{21}{20} \times \frac{21}{20}}$

⇒ $\frac{370440 \times 20 \times 20 \times 20}{21 \times 21 \times 21} = 40 \times 20 \times 20 \times 20$

= 32000. A

POSTAL A B C

45 Ans Let certain number of men = x.

Total work done by x men = $60x$.

IInd condition

Time taken by $(x+8)$ men to complete work

$= 60-10 = 50$ days.

According to question

$60x = (x+8)50$

$\Rightarrow 60x - 50x = 400$

and $x = 40$ Ans.

46 Ans Let daughter's age = x

Then mother's age = $x+30$

and $x + (x+30) = 60$

$\Rightarrow 2x = 30$

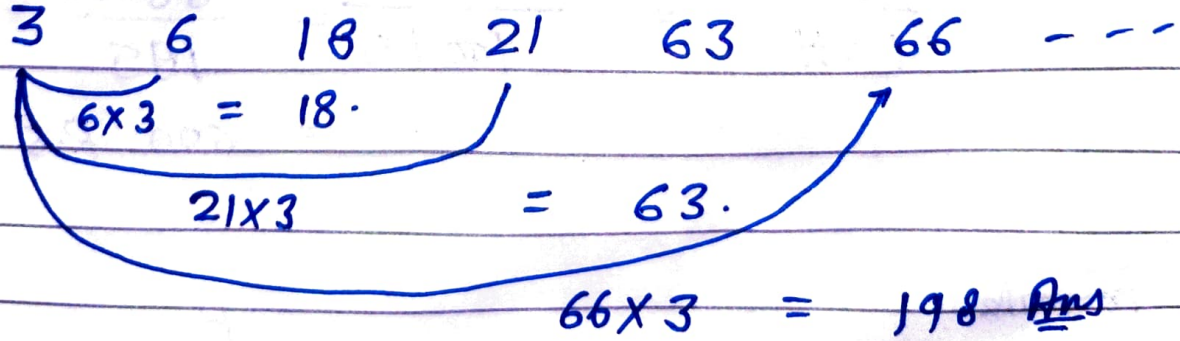
and $x = 15$.

So age of mother = $x+30$

$= 15+30 = 45$ years.

47 Ans 3, 6, 18, 21, 63, 66 - ----

Ans - 198.



POSTAL A B C

48 Female Officers = 5

Take part in Swachhata Abhiyan = $\frac{1}{5} \times 5 = 1$

Male Officers = 8

Take part in swachhata Abhiyan = $\frac{1}{8} \times 8 = 1$

Part of officers took part in comp = $\frac{1+1}{5+8} = \frac{2}{13}$ Ans.

49 Ans Price after Ist discount

$= 1000 - 1000 \times \frac{20}{100} = 1000 - 200 = 800$

Price after IInd discount

$= 800 - 10\%$ of $800 = 800 - \frac{10}{100} \times 800$

$= 800 - 80 = 720$ R.S.

Price after IIIrd discount

$= 720 - 5\%$ of $720 = 720 - \frac{5}{100} \times 720$

$= 720 - 36 = 684$ R.S.

Now SP = 684 Profit = 14%.

$CP = \frac{SP \times 100}{100 + P\%} = \frac{684 \times 100}{100 + 14} = \frac{68400}{114}$

$= 600$ R.S.

50 Ans Reduction % in consumption

$= \left(\frac{r}{100 + r} \times 100\right)\%$

$= \frac{25}{125} \times 100 = 20\%$.

POSTAL A B C